

Turn rows into visual answers.

Today we use AI as a data thinking partner, but the hands-on tool is simple: Google Sheets. No Python, no software setup, no coding environment.

Google Sheets

Chat tools

Chart choice

Data cleaning

Practical workshop

Final output

A cleaned mini dataset, 4 useful charts, and a one-page visual insight summary.

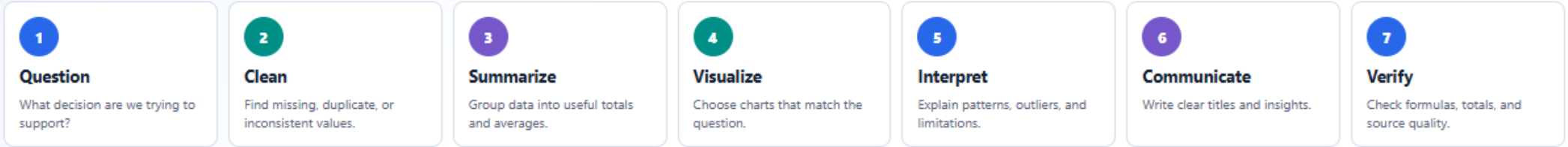
Skill stack

1. Ask better questions from data.
2. Clean messy spreadsheet columns.
3. Choose the right chart for the message.
4. Use AI to explain patterns and spot risks.
5. Design charts that people can read quickly.

Hands-on outputs

1. One sample dataset in Google Sheets.
2. Pivot table or summary table.
3. Bar chart, line chart, pie/donut alternative, and scatter or histogram.
4. Three written insights with evidence.
5. One chart repair checklist.

AI does not replace judgment. It speeds up the questions.



● ● ● TYPES OF DATA VISUALIZATIONS

Comparison

Bar, column, grouped bar. Best for comparing categories like departments, products, regions, or classes.

Trend over time

Line, area, sparkline. Best for months, weeks, dates, growth, decline, and seasonality.

Part-to-whole

Stacked bar, 100% stacked bar, treemap, donut. Best for share, composition, and mix.

Relationship

Scatter plot, bubble chart. Best for asking whether two numbers move together.

Distribution

Histogram, box plot, frequency table. Best for spread, typical values, and unusual values.

Location

Map chart. Best when geography is central to the question and location names are clean.

● ● ● CHART CHOICE MATRIX

Question	Use this chart	Avoid this mistake
Which category is biggest?	Sorted bar chart	Using pie with many slices
How did it change over time?	Line chart	Using random category order
What percent belongs to each group?	100% stacked bar or donut	Comparing tiny slices visually
Are two numbers related?	Scatter plot	Assuming correlation proves cause
Where are values concentrated?	Histogram	Only showing the average
Which values need attention?	Table with conditional formatting	Hiding exact numbers in a fancy chart

Rule: one chart should answer one question

Clean data shape

1. One row equals one record.
2. One column equals one variable.
3. Headers are clear and unique.
4. Dates are actual dates.
5. Numbers are numbers, not text.
6. Categories use consistent spelling.

CLEANING AUDIT PROMPT I have a Google Sheets dataset with these columns:

[PASTE COLUMN NAMES]

Act as a careful data analyst. Tell me:

1. Which columns are dimensions and which are metrics.
2. What data quality problems I should check.
3. Which columns may need standard spelling or formatting.
4. What summary tables and charts are possible.
5. What questions this dataset can and cannot answer.

Do not invent values. Ask me for sample rows if needed.

● ● ● **CLASS DATASET | COPY INTO GOOGLE SHEETS**

CSV SAMPLE Date,Channel,Region,Product,Leads,Sales,Revenue,Rating

```
2026-05-01,Instagram,North,Starter,42,8,960,4.2
2026-05-02,Facebook,West,Starter,35,6,720,4.1
2026-05-03,Search,East,Pro,61,14,2800,4.6
2026-05-04,Referral,South,Pro,28,9,1800,4.8
2026-05-05,Instagram,East,Starter,47,10,1200,4.3
2026-05-06,Search,North,Pro,72,17,3400,4.7
2026-05-07,Facebook,South,Basic,30,5,400,3.9
2026-05-08,Referral,West,Pro,24,8,1600,4.6
2026-05-09,Instagram,South,Basic,54,7,560,4.0
2026-05-10,Search,West,Starter,66,12,1440,4.4
2026-05-11,Facebook,East,Basic,27,4,320,3.8
2026-05-12,Referral,North,Pro,31,11,2200,4.9
```

How to use it

1. Copy the CSV text.
2. Open Google Sheets.
3. Paste into cell A1.
4. Use Data > Split text to columns if needed.
5. Format Date as date and numeric columns as numbers.

Copy CSV

DATASET TOPIC

marketing leads and sales

AUDIENCE

business owner

DECISION

which channel deserves more attentio

TOOL

Google Sheets

COLUMNS

Date, Channel, Region, Product, Leads, Sales, Revenue, Rating

Build analysis prompt

Act as a beginner-friendly data analyst and visualization coach.

Dataset topic:
marketing leads and sales

Audience:
business owner

Decision to support:
which channel deserves more attention

Tool:
Google Sheets

Columns:
Date, Channel, Region, Product, Leads, Sales, Revenue, Rating

Help me analyze this dataset without Python or coding.

Return:
1. The most important questions this data can answer.
2. Which columns are dimensions and which are metrics.
3. Data quality checks I should do in Google Sheets.
4. Four useful summary tables or pivot tables.

Copy generated prompt

Use Google Sheets as the calculation engine.

1

Paste data

Start in cell A1.

2

Format

Fix dates and numbers.

3

Filter

Check categories and blanks.

4

Summarize

Insert pivot table or use functions.

5

Chart

Insert chart from summary.

6

Title

Write the insight as the title.

7

Check

Compare chart to table totals.

Time: 25 minutes

No code required

Pivot table logic

A pivot table answers: group by this category, calculate this number, then compare the result.

Rows = categories

Values = numbers

Filters = focus

A pivot table is just a question with groupings

PIVOT PLANNING PROMPT Given these columns:

[PASTE COLUMNS]

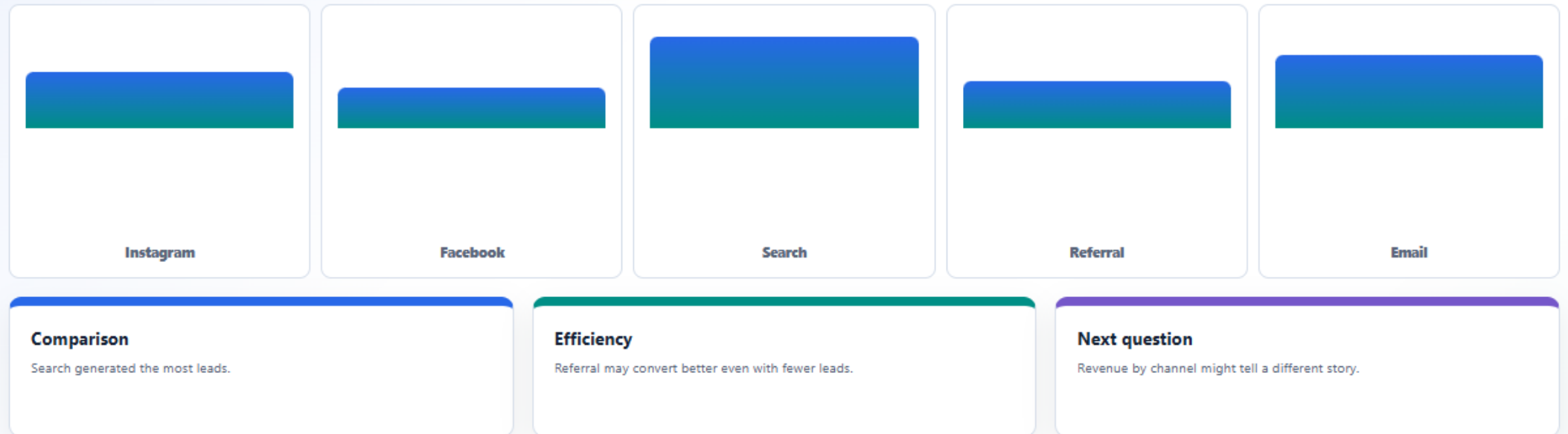
Suggest 5 beginner-friendly pivot tables for Google Sheets.

For each one, specify:

- Rows field
- Columns field, if useful
- Values field and calculation
- Filter field, if useful
- Best chart type
- The business or classroom question it answers

Keep the suggestions simple enough for beginners.

● ● ● VISUAL EXAMPLE | SAME DATA, DIFFERENT QUESTIONS



Weak vs strong titles

Weak	Stronger
Revenue by Channel	Search brings the highest revenue in May
Sales Trend	Sales rise after May 6, led by Pro product
Region Chart	North and East outperform South in conversion

The title should tell viewers what to notice

INSIGHT TITLE PROMPT Rewrite these chart titles so each title states the main insight, not just the chart topic.

Audience: [AUDIENCE]

Charts and observations:
[PASTE CHART TOPIC + WHAT YOU SEE]

Rules:

- Use plain language.
- Include the metric and category when helpful.
- Avoid exaggeration.
- Mention uncertainty if the dataset is small.

● ● ● **USE AI TO INTERPRET, NOT INVENT**

INSIGHT EXPLANATION PROMPT I created these summary results in Google Sheets:

[PASTE SUMMARY TABLE]

Act as a cautious data analyst. Write:

1. Three insights supported by the numbers.
2. One possible outlier or unusual pattern.
3. Two follow-up questions we should ask.
4. One limitation of this dataset.
5. A plain-language recommendation for [AUDIENCE].

Rules:

- Use only the numbers I provide.
- Do not invent causes.
- If a relationship is only correlation, say so.
- Keep the explanation beginner-friendly.

Good AI use

- Ask for patterns.
- Ask for plain-language explanations.
- Ask for chart suggestions.
- Ask for limitations.
- Verify every number in the sheet.

Copy insight prompt

● ● ● CHART DESIGN RULES

Sort intentionally

Sort bars descending for ranking, or use natural order for time and stages.

Reduce clutter

Remove heavy borders, unnecessary legends, and decorative 3D effects.

Use color meaningfully

Use one highlight color to direct attention; avoid rainbow charts.

Label clearly

Use readable axis labels, units, and direct labels when useful.

Show context

Include comparison, baseline, date range, or sample size when needed.

Be honest

Do not crop axes or choose chart types that exaggerate small differences.

● ● ● REPAIR BAD CHARTS FAST

CHART REPAIR PROMPT Audit this chart description:

[DESCRIBE OR PASTE THE CHART DETAILS]

Audience:

[AUDIENCE]

Purpose:

[DECISION OR QUESTION]

Tell me:

1. Whether this is the right chart type.
2. What should be sorted, filtered, or grouped.
3. What title would make the insight clearer.
4. What labels, units, or source notes are missing.
5. What visual clutter should be removed.
6. Whether the chart could mislead viewers.

Give a repaired chart plan for Google Sheets.

Common repairs

- Replace pie with sorted bar.
- Use line chart for dates.
- Move text into the title.
- Show percentages and counts.
- Combine tiny categories as Other.
- Add source and date range.

Copy repair prompt

Build in Google Sheets

1. Create a summary table by Channel.
2. Create a revenue chart by Channel.
3. Create a sales trend by Date.
4. Create a rating comparison by Product.
5. Write 3 insight titles.
6. Add one limitation note.

Time: 40 minutes

Submit or present

1. What chart is most useful?
2. What number supports your insight?
3. What surprised you?
4. What should be investigated next?
5. What chart did you choose not to use?

● ● ● DATA SAFETY WITH CHAT TOOLS

Do not paste private data

Avoid names, phone numbers, addresses, IDs, grades, health details, salary, or sensitive business data.

Use samples

Paste column names, fake rows, or aggregated summary tables instead of raw private records.

Mask details

Replace names with Person A, Person B; replace exact locations with regions.

Verify formulas

AI may suggest wrong formulas. Test on a small sample before trusting results.

Keep sources

Record where the data came from, date range, and what was excluded.

Be humble

Charts show patterns. They do not automatically explain causes.

 FACILITATOR VIEW | DAY 15 SCHEDULE

Time	Block	Visible output
00:00-00:20	What data visualization is and where AI helps	Question-first mindset
00:20-00:45	Chart families and chart choice matrix	Chart choice notes
00:45-01:10	Copy sample data and clean columns in Google Sheets	Clean spreadsheet
01:10-01:20	Break	Reset
01:20-02:05	Pivot tables and first charts	2 charts
02:05-02:40	AI prompts for insights, titles, and chart repair	Insight notes
02:40-02:50	Break	Reset
02:50-03:35	Mini project: one-page visual summary	4 charts plus 3 insights
03:35-04:00	Peer review and reflection	Repaired chart and exit ticket

Tools used in class

- Google Sheets for spreadsheet charts and chart types.
- Free chat tools for prompts, explanation, and review.
- Looker Studio/Data Studio comes on Day 16.

Tool features and access can change

Official references

- Google Sheets chart help: support.google.com/docs/answer/190718
- Data Studio reporting entry: datastudio.google.com/navigation/reporting
- Looker Studio docs: docs.cloud.google.com/looker/docs/studio

A good chart answers a real question clearly.

Clean data

Right chart

Insight title

Verified numbers

Plain language

Print / Save PDF

Exit ticket

1. Which chart type did you understand best?
2. Which chart did AI help improve?
3. What data quality issue did you find?
4. What insight can you support with numbers?
5. What should your dashboard show tomorrow?